

Machine Learning

Homework 3

Nov. 2025

Fill your Name
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1. Consider a multi-layer neural network with an input layer (d neurons), one hidden layer (n neurons), and an output layer (q neurons), just as Figure 1. The parameters between the input to hidden layer are $\mathbf{W}_1 \in \mathbb{R}^{d \times n}$ and $\mathbf{b}_1 \in \mathbb{R}^n$. The parameters between the hidden layer and the output layer are $\mathbf{W}_2 \in \mathbb{R}^{n \times q}$ and $\mathbf{b}_2 \in \mathbb{R}^q$. Note that $\mathbf{W}_1, \mathbf{W}_2$ are weight matrices and $\mathbf{b}_1, \mathbf{b}_2$ are bias vectors.

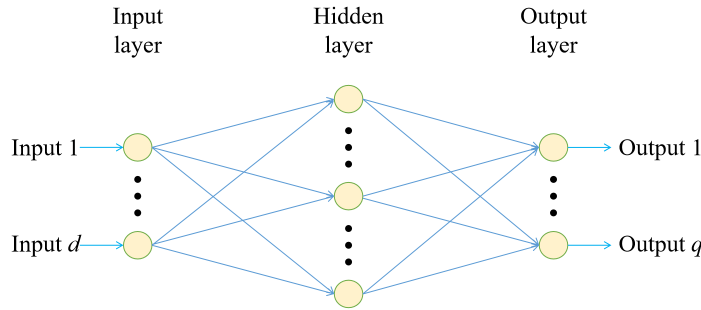


Figure 1: The diagram of neural network.

The forward propagation can be described as follows. Let $\mathbf{x} \in \mathbb{R}^d$ be the input vector. The pre-activation at the hidden layer is:

$$\mathbf{z}_1 = \mathbf{W}_1^\top \mathbf{x} + \mathbf{b}_1 \in \mathbb{R}^n.$$

Then we use the sigmoid activation function, denoted as $\sigma(\cdot)$, applied element-wise:

$$\mathbf{h} = \sigma(\mathbf{z}_1) = \frac{1}{1 + e^{-\mathbf{z}_1}} \in \mathbb{R}^n.$$

The output layer pre-activation is:

$$\mathbf{z}_2 = \mathbf{W}_2^\top \mathbf{h} + \mathbf{b}_2 \in \mathbb{R}^q,$$

and the final prediction is obtained via the softmax function:

$$\hat{\mathbf{y}} = \text{Softmax}(\mathbf{z}_2) \in \mathbb{R}^q,$$

where the k -th element is $\hat{y}_k = \frac{e^{z_{2,k}}}{\sum_{j=1}^q e^{z_{2,j}}}$.

For the loss function, we use the standard cross-entropy loss given in Equation (1). Let $\mathbf{y} \in \{0, 1\}^q$ be the one-hot encoded ground-truth vector:

$$L = - \sum_{k=1}^q y_k \ln(\hat{y}_k). \quad (1)$$

You are required to derive the gradients of the Loss function in Equation (1) with respect to all parameters, which is $\mathbf{W}_1, \mathbf{W}_2, \mathbf{b}_1, \mathbf{b}_2$. Show the intermediate steps using the chain rule. That is to say, you need to find the expression for:

$$\frac{\partial L}{\partial \mathbf{W}_2}, \quad \frac{\partial L}{\partial \mathbf{b}_2}, \quad \frac{\partial L}{\partial \mathbf{W}_1}, \quad \frac{\partial L}{\partial \mathbf{b}_1}.$$

(Hint: Recall that the derivative of the sigmoid function is $\sigma'(u) = \sigma(u)(1 - \sigma(u))$.)

Ans: For Cross-Entropy loss combined with Softmax, the derivative simplifies to the difference between prediction and ground truth:

$$\boldsymbol{\delta}_2 = \frac{\partial L}{\partial \mathbf{z}_2} = \hat{\mathbf{y}} - \mathbf{y} \in \mathbb{R}^q.$$

Using the chain rule $\frac{\partial L}{\partial \mathbf{W}_2} = \frac{\partial \mathbf{z}_2}{\partial \mathbf{W}_2} \cdot \frac{\partial L}{\partial \mathbf{z}_2}$:

$$\begin{aligned} \frac{\partial L}{\partial \mathbf{b}_2} &= \boldsymbol{\delta}_2 = \hat{\mathbf{y}} - \mathbf{y}, \\ \frac{\partial L}{\partial \mathbf{W}_2} &= \mathbf{h} \boldsymbol{\delta}_2^\top = \mathbf{h}(\hat{\mathbf{y}} - \mathbf{y})^\top. \end{aligned}$$

For the gradient w.r.t hidden layer pre-activation, we can firstly propagate the error to the hidden layer output $\mathbf{h}$ using chain rule:

$$\frac{\partial L}{\partial \mathbf{h}} = \frac{\partial \mathbf{z}_2}{\partial \mathbf{h}} \cdot \frac{\partial L}{\partial \mathbf{z}_2} = \mathbf{W}_2 \boldsymbol{\delta}_2,$$

then multiply element-wise by the derivative of the activation function. Since $\sigma'(\mathbf{z}) = \sigma(\mathbf{z})(1 - \sigma(\mathbf{z})) = h(1 - h)$, we have:

$$\delta_1 = \frac{\partial L}{\partial \mathbf{z}_1} = (\mathbf{W}_2 \delta_2) \odot [\mathbf{h} \odot (1 - \mathbf{h})].$$

Thus the gradients w.r.t input parameters are:

$$\begin{aligned} \frac{\partial L}{\partial \mathbf{b}_1} &= \delta_1, \\ \frac{\partial L}{\partial \mathbf{W}_1} &= \mathbf{x} \delta_1^\top. \end{aligned} \tag{2}$$

2. Consider a specific simple case to perform one step of Gradient Descent manually, with the input $\mathbf{x} = [1, 0]^\top$ and the ground truth $\mathbf{y} = [1, 0]^\top$. We can say the dimension of input $d = 2$, hidden $n = 2$ and output $q = 2$. The initial values of the parameters are listed below:

$$\mathbf{W}_1 = \begin{bmatrix} 0.5 & 0.2 \\ 0.5 & 0.2 \end{bmatrix} \in \mathbb{R}^{2 \times 2}, \quad \mathbf{b}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \in \mathbb{R}^2.$$

$$\mathbf{W}_2 = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \in \mathbb{R}^{2 \times 2}, \quad \mathbf{b}_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \in \mathbb{R}^2.$$

For this numerical problem (Problem 2), to make calculations easier, assume the hidden layer activation function is ReLU:

$$\mathbf{h} = \text{ReLU}(\mathbf{z}_1).$$

- Perform the forward propagation to calculate $\hat{\mathbf{y}}$.
- Perform the back-propagation to calculate the gradient matrix $\frac{\partial L}{\partial \mathbf{W}_2}$. (Give the numerical result.)
- Update $\mathbf{W}_2$ using gradient descent with a learning rate $\eta = 0.1$.

$$\mathbf{W}_2^{\text{new}} \leftarrow \mathbf{W}_2 - \eta \cdot \frac{\partial L}{\partial \mathbf{W}_2}.$$

(Note: Round your answers to **4 decimal places** where necessary)

Ans:

(a) The $\hat{\mathbf{y}}$ can be get through the following derivation:

$$\begin{aligned}\mathbf{z}_1 &= \mathbf{W}_1^\top \mathbf{x} + \mathbf{b}_1 = \begin{bmatrix} 0.5 & 0.5 \\ 0.2 & 0.2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.2 \end{bmatrix} . \\ \mathbf{h} &= \text{ReLU}(\mathbf{z}_1) = \begin{bmatrix} 0.5 \\ 0.2 \end{bmatrix} . \\ \mathbf{z}_2 &= \mathbf{W}_2^\top \mathbf{h} + \mathbf{b}_2 = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.7 \\ -0.7 \end{bmatrix} . \\ \hat{\mathbf{y}} &= \text{Softmax}(\mathbf{z}_2) = \begin{bmatrix} 0.8022 \\ 0.1978 \end{bmatrix} .\end{aligned}$$

(b) The numerical result of $\frac{\partial L}{\partial \mathbf{W}_2}$ can be get through the following derivation:

$$\begin{aligned}\boldsymbol{\delta}_2 &= \hat{\mathbf{y}} - \mathbf{y} = \begin{bmatrix} 0.8022 \\ 0.1978 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -0.1978 \\ 0.1978 \end{bmatrix} , \\ \frac{\partial L}{\partial \mathbf{W}_2} &= \mathbf{h} \boldsymbol{\delta}_2^\top = \begin{bmatrix} 0.5 \\ 0.2 \end{bmatrix} \begin{bmatrix} -0.1978 & 0.1978 \end{bmatrix} \\ &= \begin{bmatrix} 0.5(-0.1978) & 0.5(0.1978) \\ 0.2(-0.1978) & 0.2(0.1978) \end{bmatrix} = \begin{bmatrix} -0.0989 & 0.0989 \\ -0.0396 & 0.0396 \end{bmatrix} .\end{aligned}$$

(c) The update result is:

$$\begin{aligned}\mathbf{W}_2^{\text{new}} &= \mathbf{W}_2 - \eta \cdot \nabla \mathbf{W}_2 \\ &= \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} - 0.1 \begin{bmatrix} -0.0989 & 0.0989 \\ -0.0396 & 0.0396 \end{bmatrix} \\ &= \begin{bmatrix} 1.0099 & -1.0099 \\ 1.0040 & -1.0040 \end{bmatrix} .\end{aligned}\tag{3}$$